

QCE Chemistry Units 3&4

Paper 1

Student's Name: _____

Teacher's Name: _____

Time allowed

- Perusal time – 10 minutes
- Working time – 90 minutes

General instructions

- Answer all questions in this question and response booklet.
- QCAA-approved calculator permitted.
- Formula and data booklet provided.
- Planning paper will not be marked.

Section 1 (25 marks)

- 25 multiple choice questions

Section 2 (35 marks)

- 8 short response questions

Students are advised that this is a trial examination only and cannot in any way guarantee the content or the format of the 2022 QCE Chemistry Units 3&4 Written Examination.

Neap[®] Education (Neap) Trial Exams are licensed to be photocopied or placed on the school intranet and used only within the confines of the school purchasing them, for the purpose of examining that school's students only. They may not be otherwise reproduced or distributed. The copyright of Neap Trial Exams remains with Neap. No Neap Trial Exam or any part thereof is to be issued or passed on by any person to any party inclusive of other schools, non-practising teachers, coaching colleges, tutors, parents, students, publishing agencies or websites without the express written consent of Neap.

SECTION 1**Instructions**

- Choose the best answer for Questions 1–25.
- This section has 25 questions and is worth 25 marks.
- Use a 2B pencil to fill in the A, B, C or D answer bubble completely.
- If you change your mind or make a mistake, use an eraser to remove your response and fill in the new answer bubble completely.

	A	B	C	D
Example:	☒	☐	☐	☐

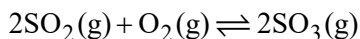
	A	B	C	D
1.	☐	☐	☐	☐
2.	☐	☐	☐	☐
3.	☐	☐	☐	☐
4.	☐	☐	☐	☐
5.	☐	☐	☐	☐
6.	☐	☐	☐	☐
7.	☐	☐	☐	☐
8.	☐	☐	☐	☐
9.	☐	☐	☐	☐
10.	☐	☐	☐	☐
11.	☐	☐	☐	☐
12.	☐	☐	☐	☐
13.	☐	☐	☐	☐
14.	☐	☐	☐	☐
15.	☐	☐	☐	☐
16.	☐	☐	☐	☐
17.	☐	☐	☐	☐
18.	☐	☐	☐	☐
19.	☐	☐	☐	☐
20.	☐	☐	☐	☐
21.	☐	☐	☐	☐
22.	☐	☐	☐	☐
23.	☐	☐	☐	☐
24.	☐	☐	☐	☐
25.	☐	☐	☐	☐

SECTION 2**Instructions**

- Write using black or blue pen.
- If you need more space for a response, use the additional pages at the back of this booklet.
 - On the additional pages, write the question number you are responding to.
 - Cancel any incorrect response by ruling a single diagonal line through your work.
 - Write the page number of your alternative/additional response, i.e. See page ...
 - If you do not do this, your original response will be marked.
- This section has eight questions and is worth 35 marks.

QUESTION 26 (2 marks)

Sulfur dioxide and oxygen are mixed to produce sulfur trioxide according to the following equation.



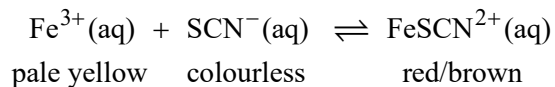
A catalyst, vanadium pentoxide, is then added to the mixture.

- a) Predict the effect that the catalyst will have on the equilibrium yield. *[1 mark]*

- b) Predict the effect that the catalyst will have on the rate of reaction. *[1 mark]*

QUESTION 27 (3 marks)

5 mL of a solution containing Fe^{3+} ions is added to a 10 mL of a solution containing SCN^- ions in a large test tube at room temperature. The reaction occurs according to the following equation.



The resulting solution is pale red.

- a) Another 5 mL of the solution containing Fe^{3+} ions is added to the test tube, and the colour of the resulting solution becomes darker.

Explain why the colour change occurred.

[1 mark]

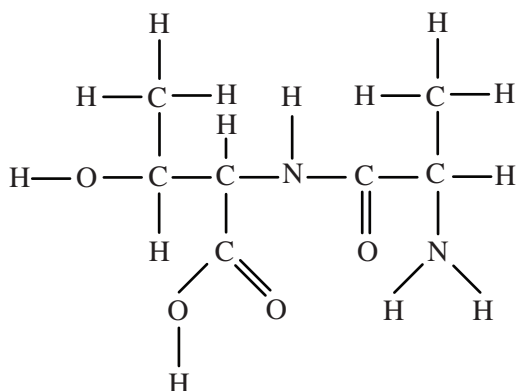
- b) The test tube from 27a) is placed in a water bath at approximately 90°C. The colour of the solution in the test tube becomes lighter, changing to yellow.

Deduce whether the reaction is endothermic or exothermic.

[2 marks]

QUESTION 29 (4 marks)

The diagram shows two amino acids that have bonded to form a dipeptide.



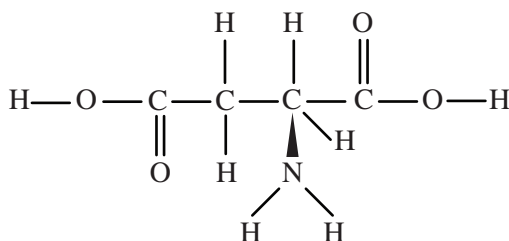
- a) Identify the two amino acids.

[2 marks]

- b) On the diagram above, circle the peptide bond that joins the amino acid molecules.

[1 mark]

- c) An amino acid is shown.

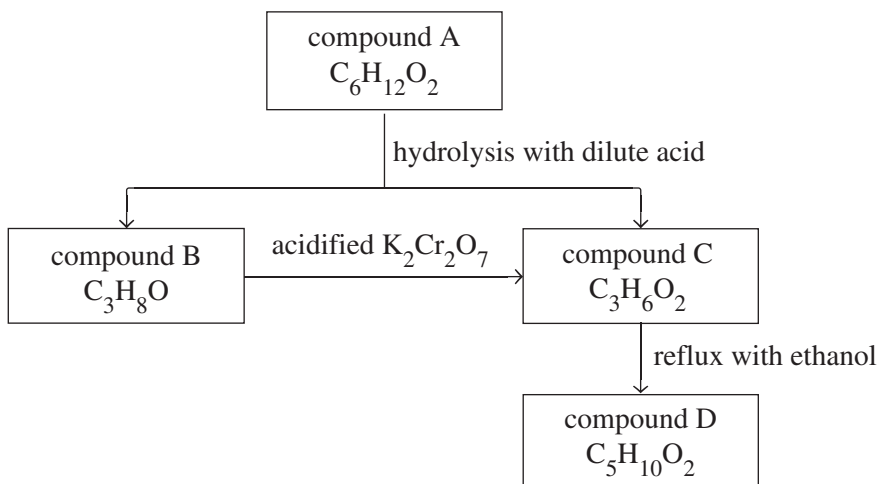


Determine why this amino acid can exist as an anion with a charge of 2-.

[1 mark]

QUESTION 30 (7 marks)

The flow chart shows a series of reactions involving four compounds: A, B, C and D.



a) Sketch the structural formulas of compounds A, C and D.

[3 marks]

Compound	Structural formula
A	
C	
D	

- b) The type of reaction that produces compound B is different to the type of reaction that produces compound D.

Identify and describe the two types of reaction.

[2 marks]

Reaction that produces compound B _____

Reaction that produces compound D _____

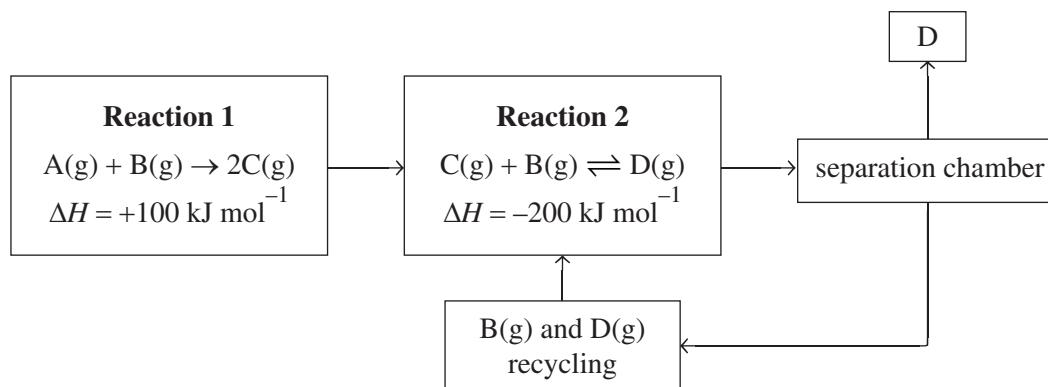
- c) Compound B has a boiling point of 97°C and compound C has a boiling point of 141.2°C.

Explain the difference in boiling points.

[2 marks]

QUESTION 31 (4 marks)

The flow chart shows the steps in a particular industrial process. The conditions for reaction 2, such as temperature and pressure, can be altered.

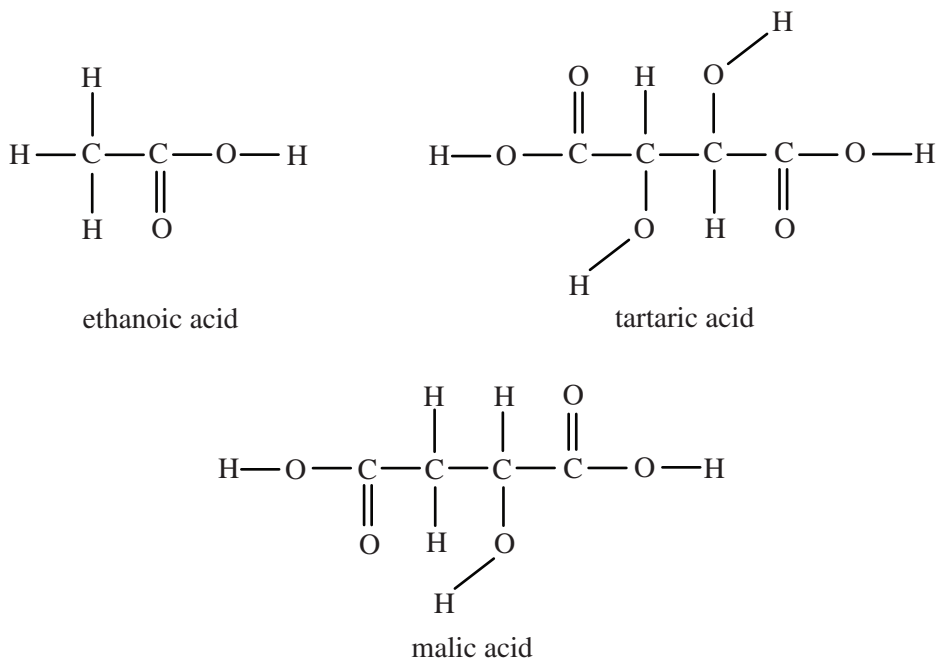


- a) If the temperature of the vessel for reaction 2 is lowered, explain the effect this will have on the yield of the reaction and the value of the equilibrium constant (K_c). [2 marks]

- b) If the pressure of the vessel for reaction 2 is increased to 5 atm by pumping more C(g) and B(g) into the vessel at constant temperature, infer the effect this will have on the yield of the reaction and the value of K_c . [2 marks]

QUESTION 32 (6 marks)

The structural formulas of three acids that are commonly found in fruit and associated with winemaking are shown.



The table shows the dissociation constants (K_a) and pK_a values of the acids.

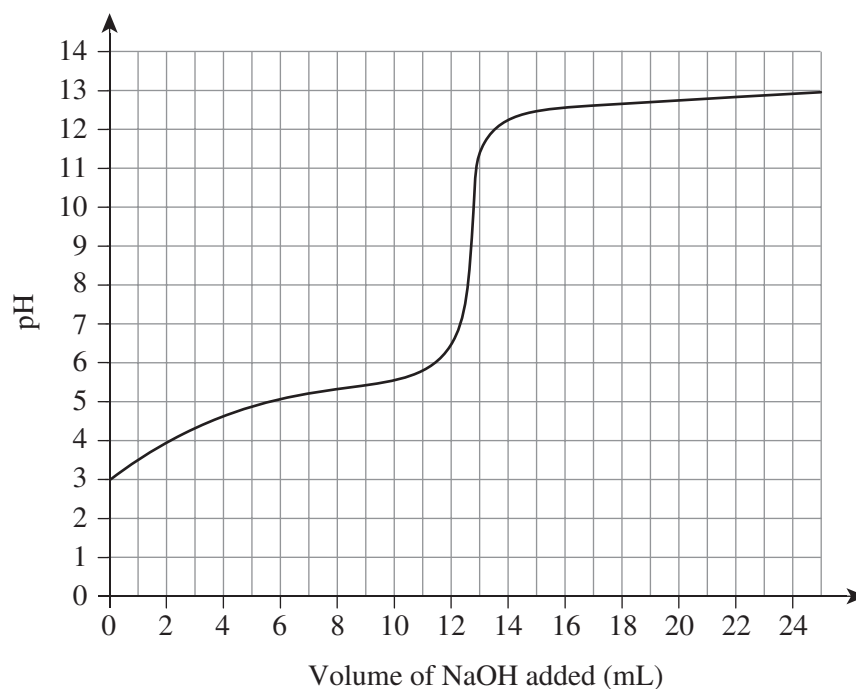
Acid	Formula	K_a	pK_a value(s)
ethanoic acid	CH_3COOH	1.75×10^{-5}	4.75
tartaric acid	$C_4H_6O_6$	1.0×10^{-3}	2.95, 4.25
malic acid	$C_4H_6O_5$	4.0×10^{-4}	3.4, 5.2

- a) Using the first pK_a value of each acid, determine which acid is the strongest. [1 mark]

- b) Propose why malic acid has two pK_a values. [1 mark]

QUESTION 33 (4 marks)

A student titrated a solution of a weak acid of unknown concentration with a 1.0 M solution of standard sodium hydroxide, NaOH. The pH of the acid solution was monitored as the titration proceeded and is shown in the graph.



- a) What was the initial pH of the acid solution? *[1 mark]*
-
- b) On the graph, label the equivalence point, half-equivalence point and buffer zone. *[3 marks]*

END OF PAPER

